DOCUMENT CONTROL & METADATA			
Project Name:	Conventional vs Toroidal Propeller Comparison	**Lead Analyst:**	AVX Engineering Team
Simulation Type:	CFD	**Software Tool:**	SolidWorks 2026
Revision:	v1.0	**Date**	Oct. 04, 2026

1. Simulation Setup

SolidWorks CFD	
Analysis Type	External
Unit System	SI (m-kg-s)
Fluid	Air
Total Cell Counts Mesh	233,564
Flow Type	Laminar and Turbulent
Gravity	On
No. of Iterations	450
Calculation Time	76 Hours
Boundary Conditions	
Velocity	0 m/s & 6 m/s
Static Pressure	101325 Pascal

SolidWorks CFD	
Angular Velocity (Rotating Region Global)	1,120.50 rad / s or 10,700 RPM

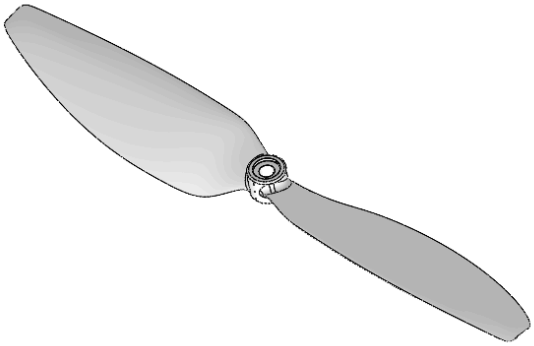


Figure 1. Conventional 6030 Propeller

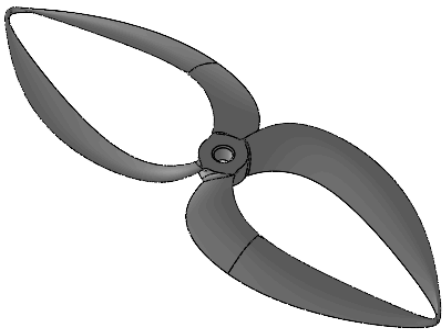


Figure 2. Toroidal Propeller

2. Simulation Report

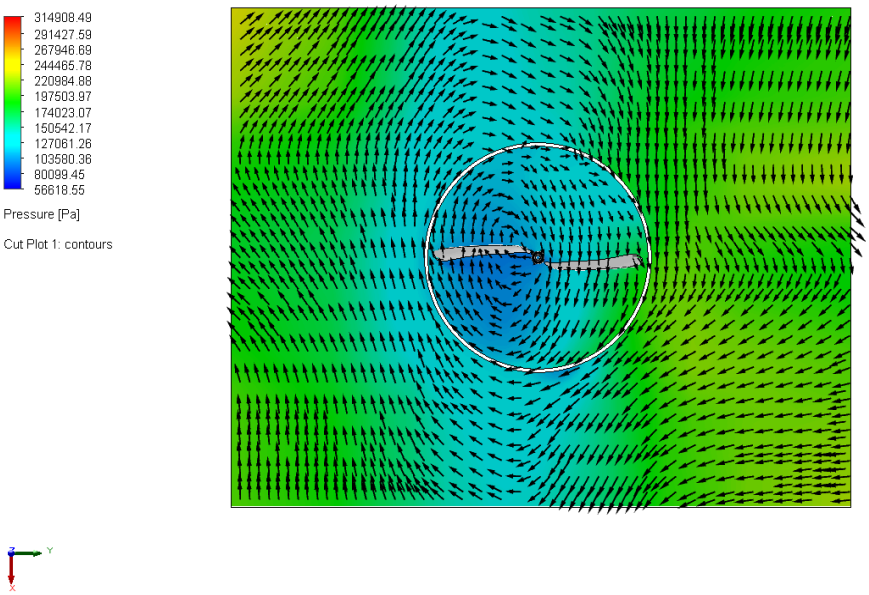


Figure 3. Conventional Pressure Cut Plot

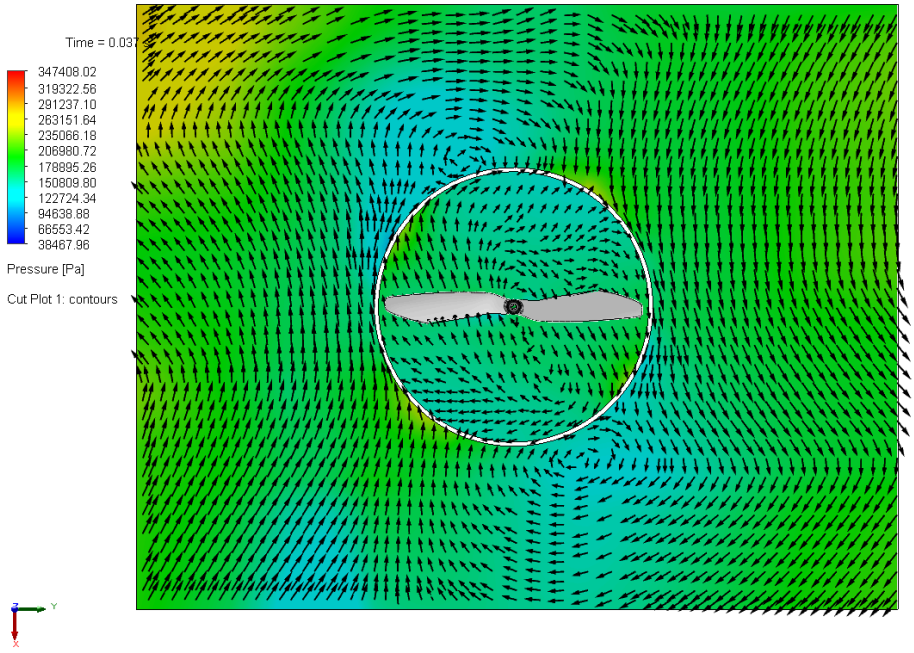


Figure 4. Conventional Pressure Cut Plot - 6 m/s

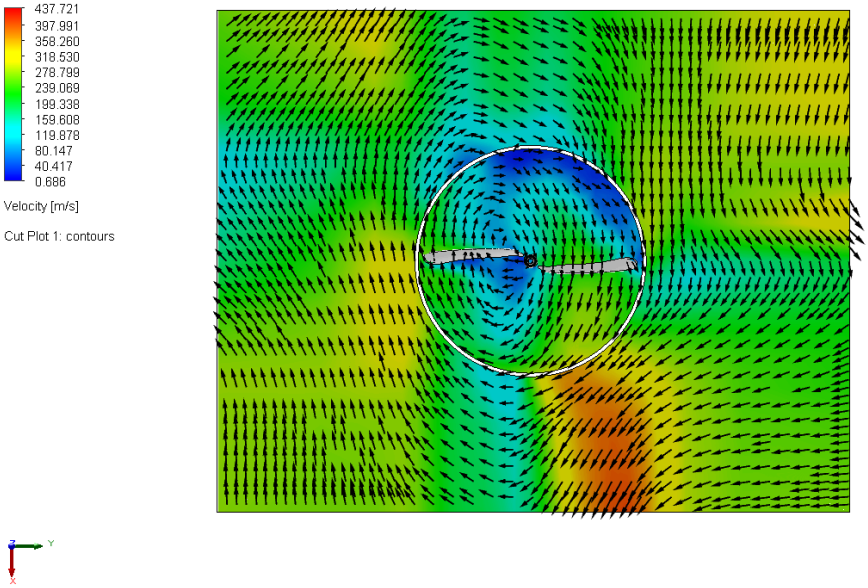


Figure 5. Conventional Velocity Cut Plot

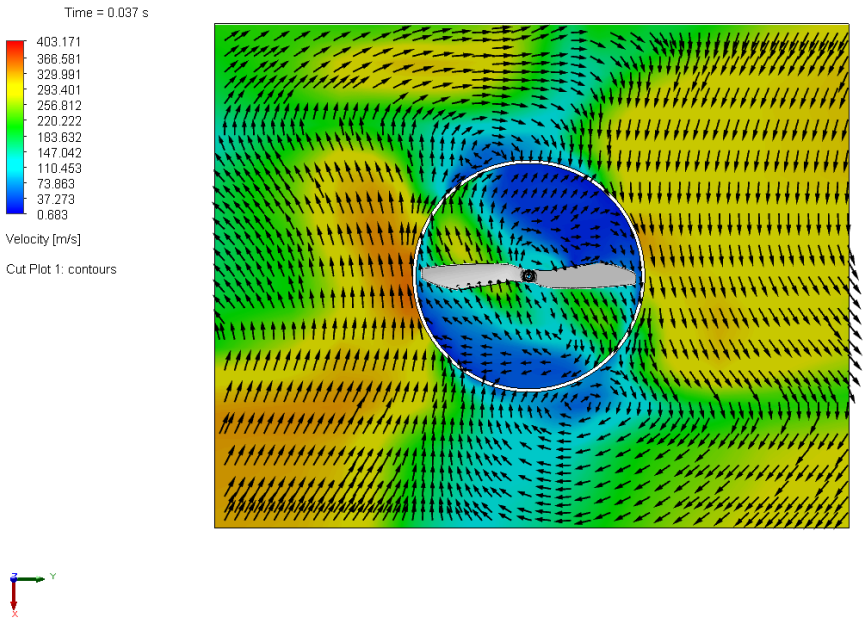


Figure 6. Conventional Velocity Cut Plot - 6 m/s

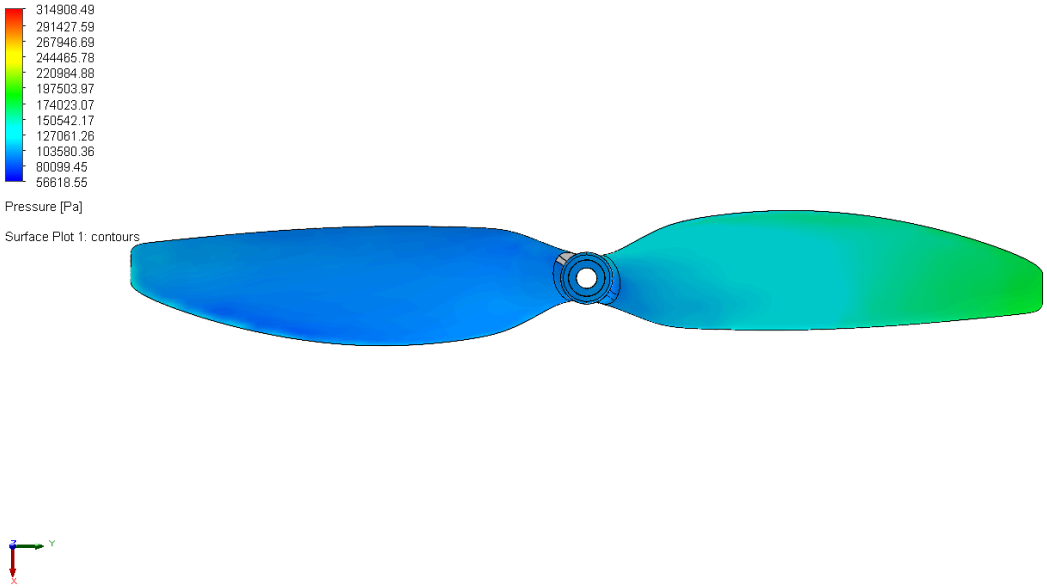


Figure 7. Conventional Surface Plot - Pressure

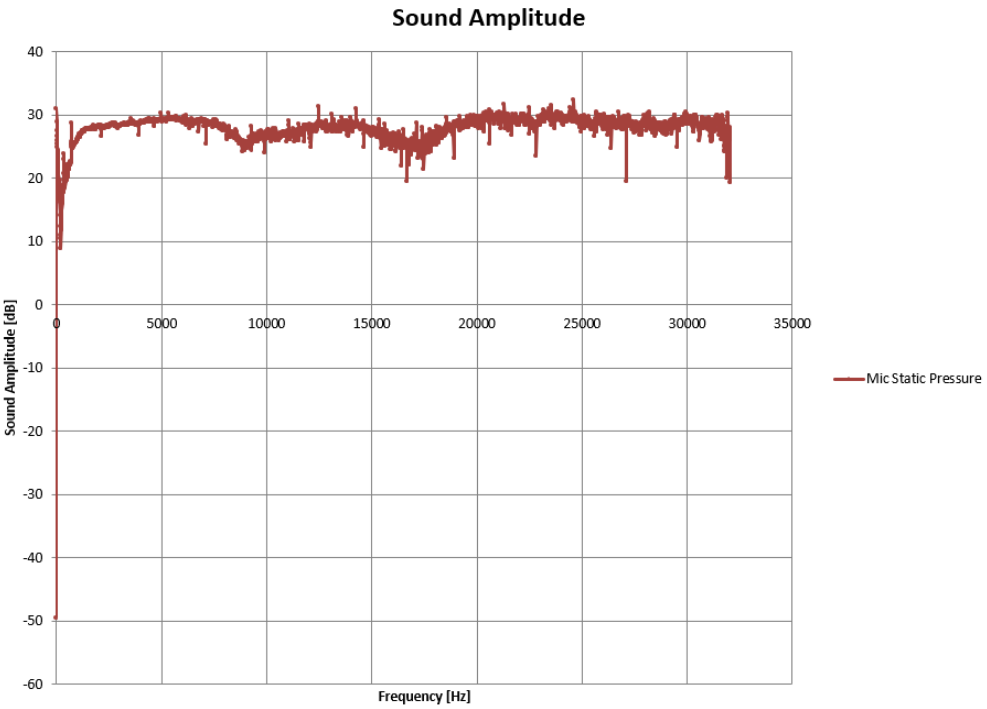


Figure 8. Conventional Sound Amplitude vs Frequency

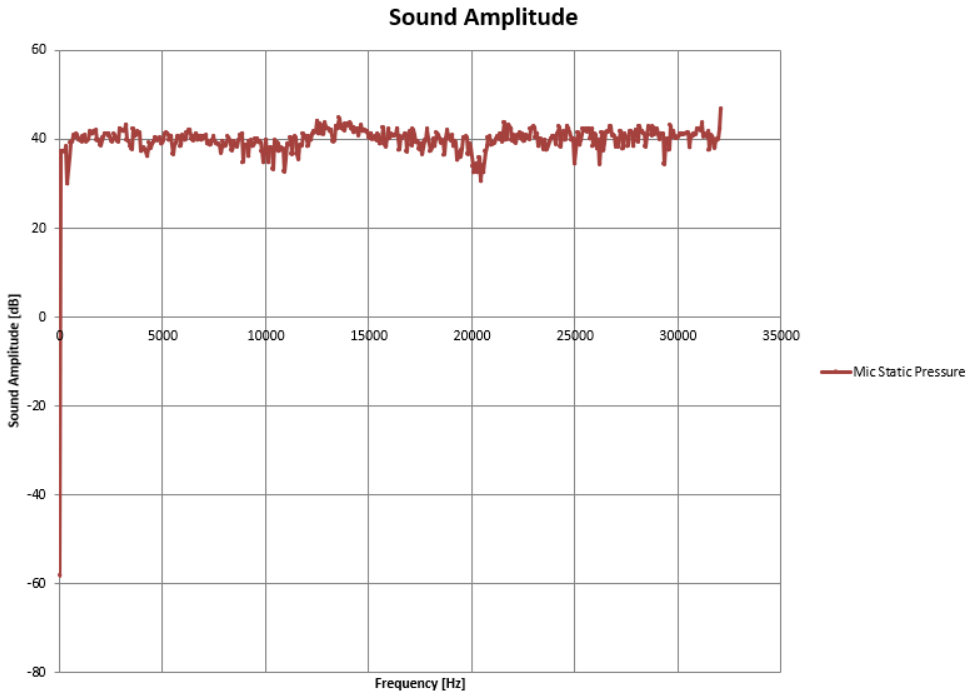


Figure 9. Conventional Sound Amplitude vs Frequency - 6m/s

Conventional Propeller Results	
Thrust @ hover	27.38 N
Torque @ hover	0.209 Nm
Power @ hover	234.2 W/m^2
Propeller Efficiency @ hover	-
Thrust @ 15 deg;6m/s	26.95 N
Torque @ 15 deg;6m/s	0.228 Nm
Power @ 15 deg;6m/s	255.3 W/m^2
Propeller Efficiency @ 15 deg;6m/s	63.5%

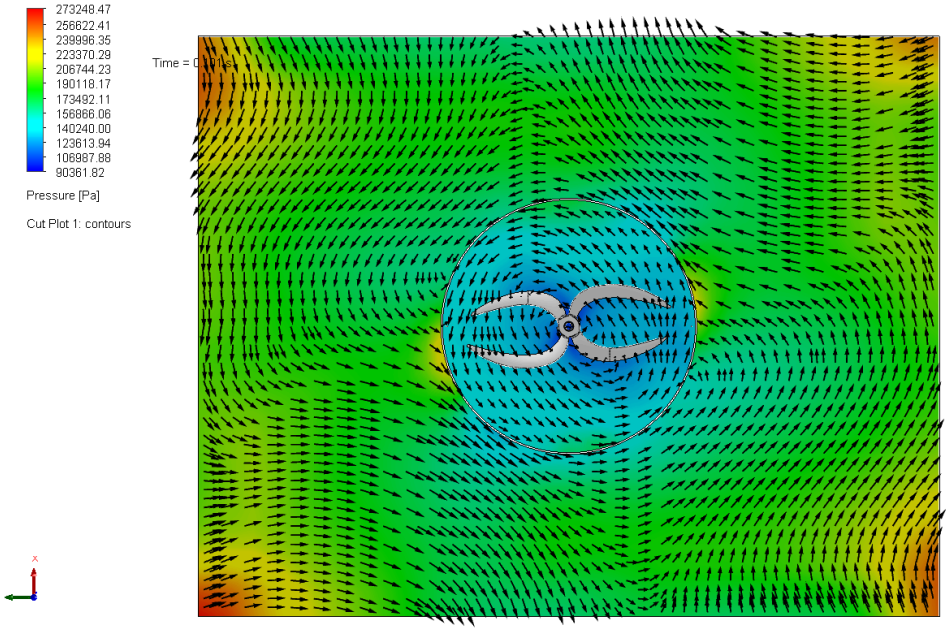


Figure 10. Toroidal Pressure Cut Plot

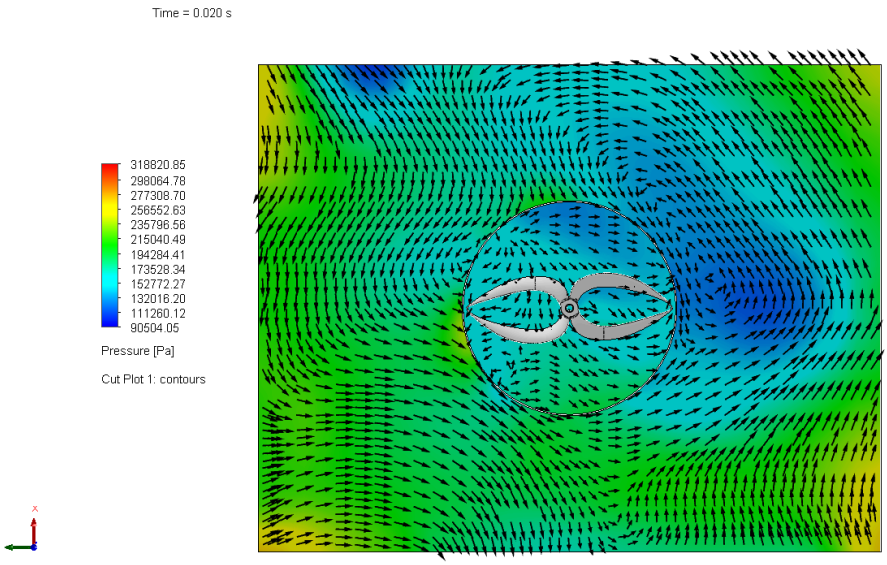


Figure 11. Toroidal Pressure Cut Plot - 6 m/s

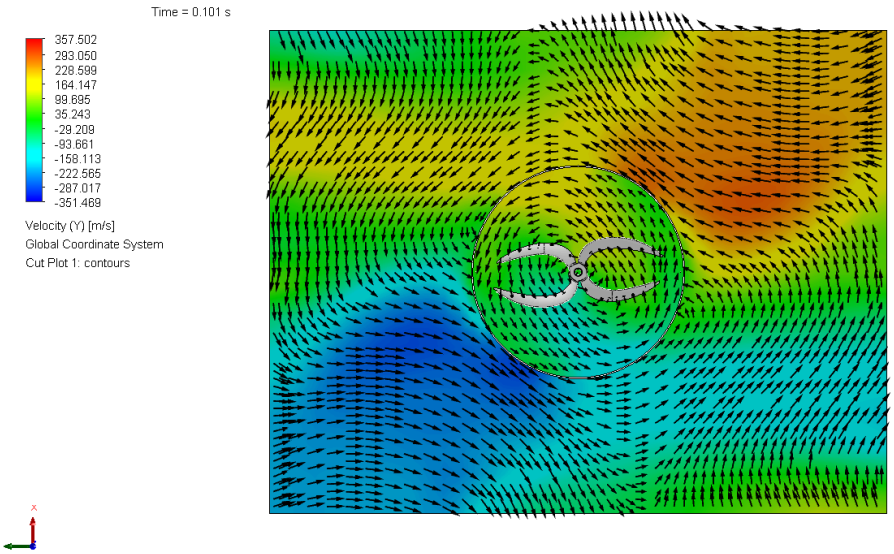


Figure 12. Toroidal Velocity Cut Plot

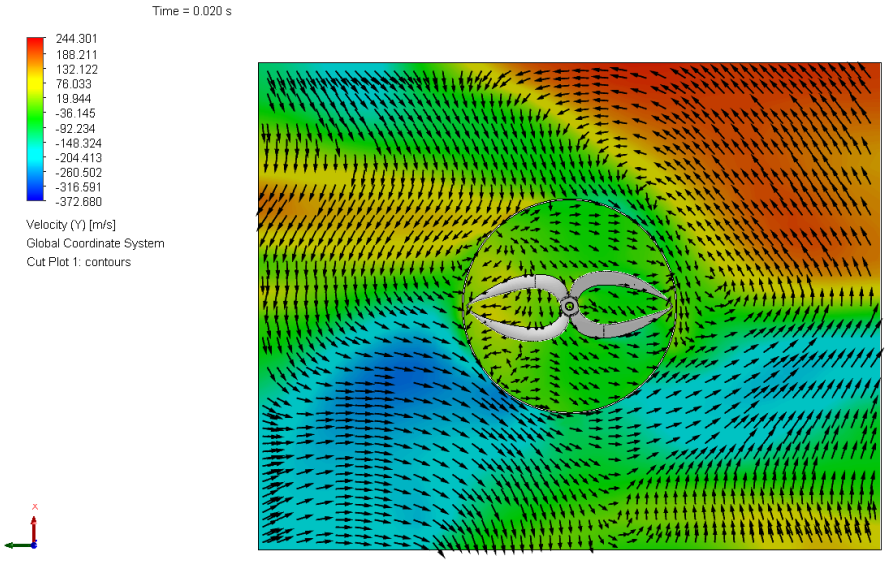


Figure 13. Toroidal Velocity Cut Plot - 6m/s

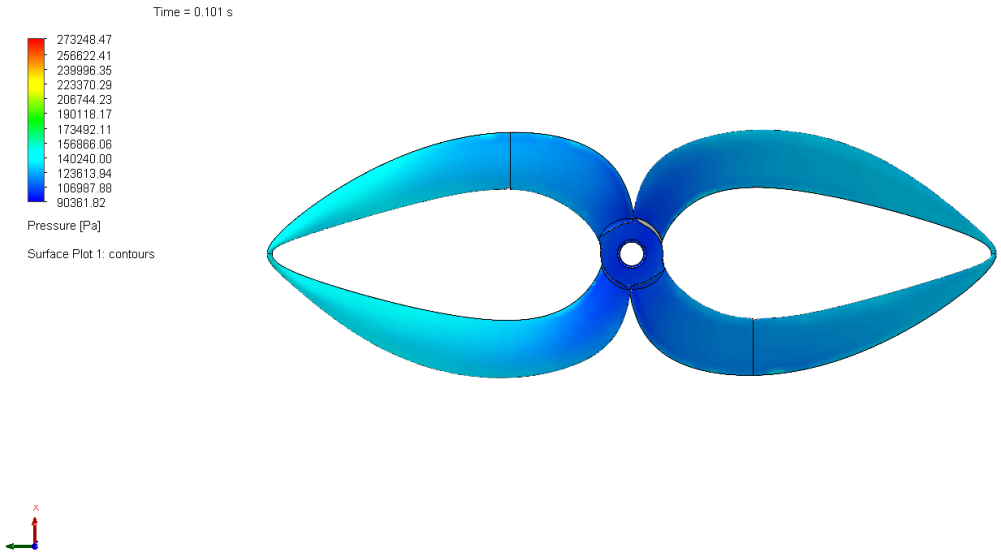


Figure 14. Toroidal Surface Plot - Pressure

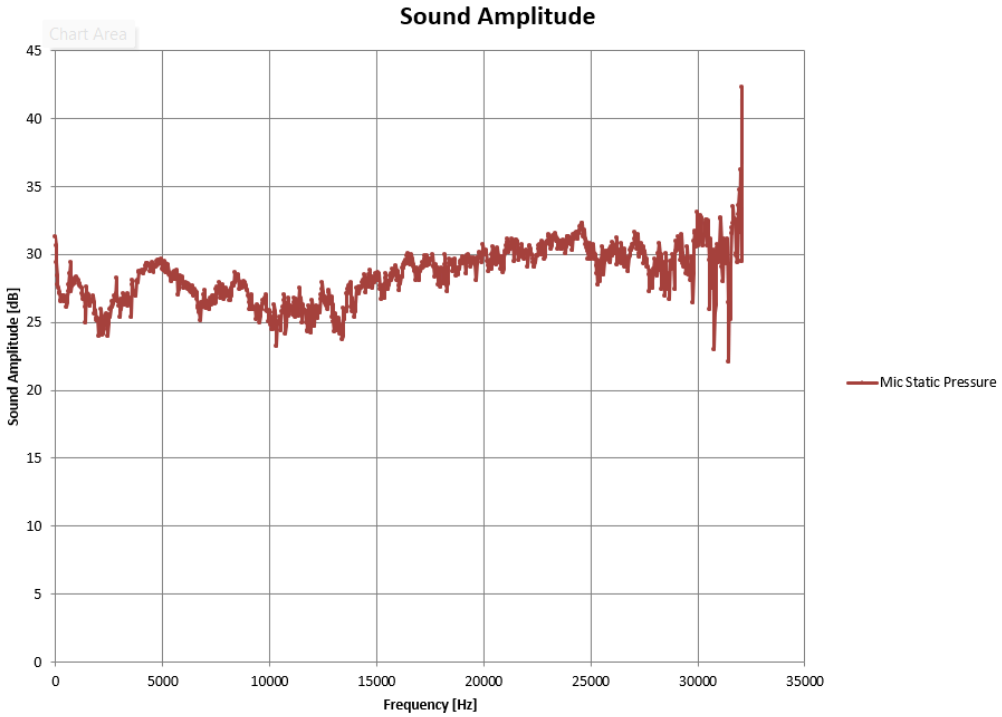


Figure 15. Toroidal Sound Amplitude vs Frequency

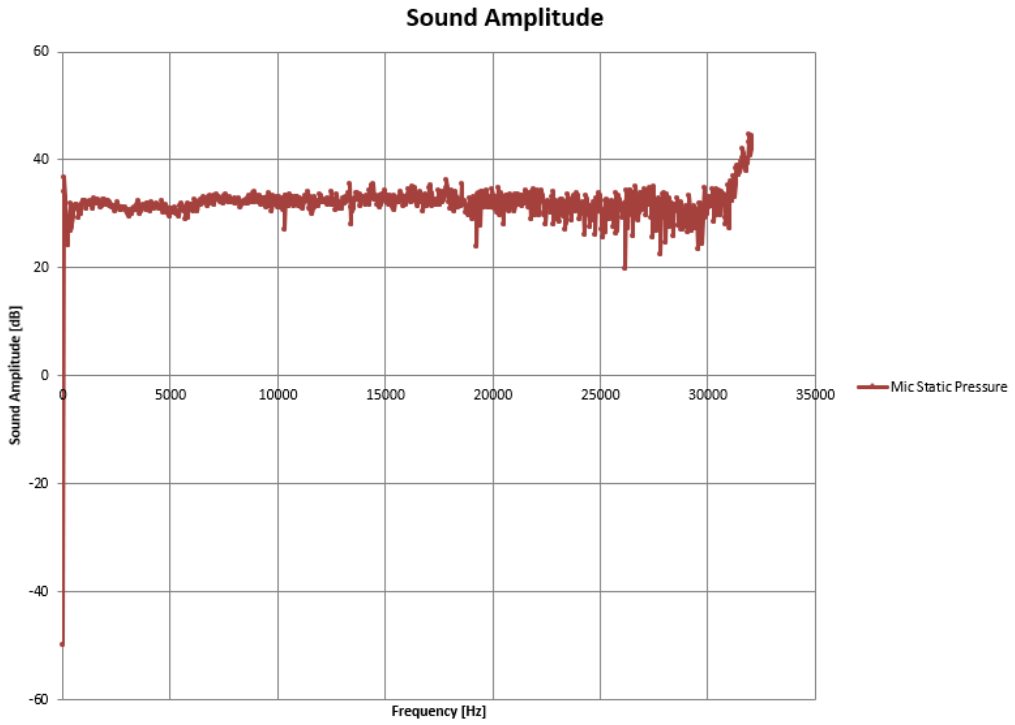


Figure 16. Toroidal Sound Amplitude vs Frequency - 6m/s

Toroidal Propeller Results	
Thrust @ hover	32.66 N
Torque @ hover	0.251 Nm
Power @ hover	281.1 W/m^2
Propeller Efficiency @ hover	-
Thrust @ 15 deg;6m/s	39.96 N
Torque @ 15 deg;6m/s	0.539 Nm
Power @ 15 deg;6m/s	603.7 W/m^2
Propeller Efficiency @ 15 deg;6m/s	40.2%

Conventional vs Toroidal Summary Results		
	Conventional	Toroidal
Thrust @ hover	27.38 N	32.66 N
Torque @ hover	0.209 Nm	0.251 Nm
Power @ hover	234.2 W/m^2	281.1 W/m^2
Propeller Efficiency @ hover	-	-
Thrust @ 15 deg;6m/s	26.95 N	39.96 N
Torque @ 15 deg;6m/s	0.228 Nm	0.539 Nm
Power @ 15 deg;6m/s	255.3 W/m^2	603.7 W/m^2
Propeller Efficiency @ 15 deg;6m/s	63.5%	40.2%

NOTHING FOLLOWS